Attention Pooling Enhances NCA-based Classification of Microscopy Images

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Abstract. Neural Cellular Automata (NCA) offer a robust and interpretable approach to image classification, making them a promising choice for microscopy image analysis. However, a performance gap remains between NCA and larger, more complex architectures. We address this challenge by integrating attention pooling with NCA to enhance feature extraction and improve classification accuracy. The attention pooling mechanism refines the focus on the most informative regions, leading to more accurate predictions. We evaluate our method on eight diverse microscopy image datasets and demonstrate that our approach significantly outperforms existing NCA methods while remaining parameter-efficient and explainable. Furthermore, we compare our method with traditional lightweight convolutional neural network and vision transformer architectures, showing improved performance while maintaining a significantly lower parameter count. Our results highlight the potential of NCA-based models an alternative for explainable image classification.

Keywords: Neural Cellular Automata · Attention Pooling · Classification.

1 Introduction

The accurate classification of cells in microscopy images is essential for medical diagnostics, aiding in the diagnosis and monitoring of various diseases. Traditionally, this task relies on medical experts, but manual analysis is costly, time-consuming, and susceptible to human error. Deep learning has significantly advanced this field, promising to provide an efficient, consistent, scalable, and accessible solution. [6,8,11,13,18,20,21,29]

However, state-of-the-art architectures such as Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) often have large parameter counts

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and require extensive training data to achieve high performance. In low-data medical imaging settings, these models are prone to overfitting and may struggle to generalize across domains. While lightweight variants have been proposed [7,14], they often trade off accuracy for reduced complexity, leaving a gap between model capacity and generalization.

A promising alternative to overcome these limitations are Neural Cellular Automata (NCA) [15]. NCA iteratively update feature maps through local interactions, enabling information propagation over multiple steps. This iterative process allows for effective feature extraction while maintaining a minimalistic design.

A key limitation in previous NCA-based classification models is the transition from high-dimensional feature maps to compact feature embeddings. Deutges et al. [4] proposed an NCA model for classification, where a simple maximum pooling operation is used to reduce dimensionality. However, this approach can result in the loss of valuable information. Tesfaldet et al. [24] explored the integration of self-attention mechanisms within an NCA framework for denoising, showing the potential of spatial attention in NCA-based models.

Here, we introduce attentionNCA (aNCA), a novel NCA-based image classification model that integrates attention pooling to enhance feature aggregation. In contrast to the simple maximum pooling proposed by Deutges et al. [4] our approach enhances NCA with a spatial attention pooling mechanism. This learnable, weighted aggregation method focuses on the most informative regions of the feature maps, improving representation quality and classification performance while maintaining a lightweight architecture with only 89k parameters. In contrast to the work of Tesfaldet et al. [24], our method differs significantly in the way the attention mechanism is integrated and specifically focuses on enhancing feature aggregation for classification tasks.

We evaluate aNCA across eight diverse microscopy datasets, including white blood cells, Pap smears, urine sediments, malaria infected cells, and pathology tissues. Our model consistently outperforms existing NCA-based classification models and conventional lightweight architectures. To foster reproducible research, we publish our source code at https://github.com/marrlab/aNCA.

2 Methods

We propose a classification network that utilizes an NCA backbone for feature extraction. To aggregate the extracted features, we introduce a learnable spatial attention mechanism that emphasizes the most informative regions. The highest 10% of pixel values across each feature channel are then averaged to produce a compact, low-dimensional feature embedding. This embedding is passed through a fully connected classifier to generate the final prediction.

2.1 NCA backbone

NCA operate through iterative updates, where each cell's state is modified based on its local neighborhood [4]. Here, a cell refers to an individual pixel across all feature channels.

The update rule for a given cell c is defined by a transition function that takes the cell's local 3×3 neighborhood N_c as input and computes an update. Firstly, information from the local neighborhood is aggregated by the perception function f_p, which applies two convolutional filters (κ_1 and κ_2) to the neighborhood. The resulting features are concatenated with the cell's current state c^t to form the perception vector $(f_p(N_c))$. The update function f_u then processes this information through a two-layer fully connected network with Rectified Linear Unit (ReLU) activations, parameterized by weights $\tilde{W}_1$, $\tilde{W}_2$ and biases $\tilde{b}_1$, $\tilde{b}_2$:

$$f_p : \mathbf{R}^{3\times 3\times n} \to \mathbf{R}^{3n}, \qquad N_c \mapsto (c, N_c * \kappa_1, N_c * \kappa_2)^T, \tag{1}$$

$$f_u : \mathbf{R}^{3n} \to \mathbf{R}^{n}, \qquad f_p(N_c) \mapsto \tilde{W}_2 \max(\tilde{W}_1 f_p(N_c) + \tilde{b}_1, 0) + \tilde{b}_2. \tag{2}$$

Finally, the output of the update function is added to the current state:

$$c^{t+1} = c^t + \delta f_u(f_p(N_c)). \tag{3}$$

On average, only half of the cells are updated per iteration, due to the stochastic binary variable δ, which acts as a regularization mechanism, enhancing robustness.

At a given time step t, the full set of cell states is represented as $S_t \in \mathbf{R}^{H\times W\times n}$ for an image with height H, width W, and n feature channels. A single NCA update across the entire image can therefore be expressed as

$$\mathrm{NCA}_\phi : \mathbf{R}^{H\times W\times n} \to \mathbf{R}^{H\times W\times n}, \tag{4}$$

where $\phi = (\kappa_1, \kappa_2, \tilde{W}_1, \tilde{W}_2, \tilde{b}_1, \tilde{b}_2)$ denotes the set of trainable parameters.

2.2 Classification Architecture

After extracting feature channels using NCA, we aggregate each channel into a single value to form a feature embedding for classification.

$$\mathrm{pool} : \mathbf{R}^{H\times W\times n} \to \mathbf{R}^{n} \tag{5}$$

To ensure a smooth transition between high and low dimensions, we implement attention pooling, which enables effective dimensionality reduction by learning an optimal weighting of the spatial features thus preserving crucial information.

Attention Pooling We define a matrix of learnable weights $\theta \in \mathbf{R}^{H\times W}$, which is used to pool each of the feature channels S^i into a weighted sum

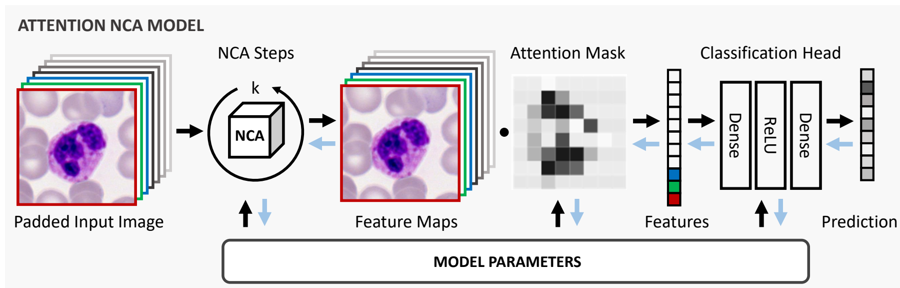

Fig. 1. Architecture of our attention neural cellular automata (aNCA). **A:** The model extracts feature maps from the padded input image using neural cellular automata. Each feature map is aggregated into a single value via a learnable weighted sum, a mechanism we call attention pooling, which enhances focus on the most informative regions. The resulting feature embedding is then passed through a fully connected network for classification. The gradient flow is indicated with blue arrows. The model including the attention mask and NCA is trained end-to-end.

$$\text{pool} : \mathbf{R}^{H\times W\times n} \to \mathbf{R}^n, \begin{pmatrix} S^1 \\ S^2 \\ \vdots \\ S^n \end{pmatrix} \mapsto \frac{1}{H\cdot W} \begin{pmatrix} \Sigma_{i,j\in H,W} S^1_{i,j}\cdot\sigma(\theta_{i,j}) \\ \Sigma_{i,j\in H,W} S^2_{i,j}\cdot\sigma(\theta_{i,j}) \\ \vdots \\ \Sigma_{i,j\in H,W} S^n_{i,j}\cdot\sigma(\theta_{i,j}) \end{pmatrix}, \tag{6}$$

where $\sigma(\theta_{i,j}) = (1+e^{-\theta_{i,j}})^{-1}$. In order to reduce noise in the pooling operation, we only consider a certain percentage of the highest activations in the weighted sum. We achieve this by sorting the products $S_{i,j}\cdot\sigma(\theta_{i,j})$, and averaging only the highest 10% of values for each feature channel. This mechanism learns the most informative regions for classification.

Classification Head To obtain the final class prediction, the aggregated feature embedding v is passed through a two-layer fully connected network

$$g : \mathbf{R}^n \to \mathbf{R}^C, \quad g(v) = \sigma(W_2 \max(W_1 v + b_1, 0) + b_2), \tag{7}$$

where W_1, W_2, b_1, b_2 are the weights and biases and σ is a softmax activation.

The learnable parameters of the model include the NCA parameters, the attention pooling weights, and the weights of the fully connected classification head. An overview of our method is illustrated in Figure 1.

3 Experiments & Results

3.1 Datasets

Eight diverse microscopy datasets are used for evaluation of our proposed method. Most of these datasets are publicly accessible. Example images from all datasets are displayed in Figure 2.

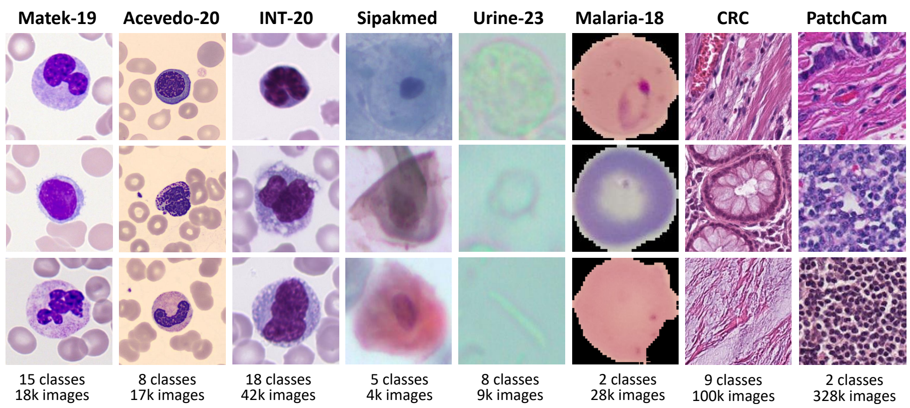

Fig. 2. Example images from the eight diverse microscopy datasets used in this study. The datasets cover a wide range of domains and differ in class distribution and size.

Matek-19 [13] consists of 18,365 expert-labeled white blood cell images sorted into 15 different classes. The samples were collected from 200 healthy donors and patients with acute myeloid leukemia at the Munich University Hospital. The images were captured at a resolution of 400×400 pixels, corresponding to a physical size of 29×29 micrometers.

Acevedo-20 [2] contains 17,092 single-cell images sourced from healthy donors at the Hospital Clinic of Barcelona. The dataset consists of 8 cell classes, with each image having a resolution of 360×363 pixels, corresponding to a scale of 36×36.3 micrometers.

INT-20 is an in-house dataset that contains approximately 42,000 single-cell images from 18 different classes. Each image has a resolution of 288×288 pixels, representing a field of view of 25×25 micrometers.

Sipakmed [16] consists of 4,049 manually annotated cell images extracted from Pap smear slides. The dataset contains 5 classes based on cytomorphology. The images were acquired using an optical microscope equipped with a CCD camera.

Urine-23 [25] comprises 8,509 labeled images containing 8 classes of urine sediment particles. The samples were collected from 409 patients at the Biochemistry Clinics of Elazig Fethi Sekin Central Hospital, captured with an Optika B293PLi microscope.

Malaria-18 [17] contains 27,560 Giemsa stained red blood cell images sorted into parasitized and uninfected. The images were obtained from 193 patients at Chittagong Medical College Hospital, Bangladesh.

CRC [9] is a collection of 100,000 image patches derived from histology slides of colorectal tissue, covering 9 classes of normal and cancerous samples. Each patch is 224×224 pixels, with a spatial resolution of 0.5 micrometers per pixel. To ensure consistency, images are color-normalized using Macenko's method[12]. An additional set of 7,180 independent samples is used for validation.

PatchCam [3] contains 327,680 histopathology image patches extracted from lymph node tissue sections, divided into 262,144 training samples, and validation and test set both of 32,768 samples. Each sample has 96×96 pixels and a binary label indicating the presence or absence of metastatic tissue.

3.2 Implementation details

In this section, we describe the training setups and implementation settings of aNCA and baselines.

Input format 3-channels RGB images, resized to 64×64 pixels and normalized with dataset-specific mean and standard deviation. For training, the images are augmented by random rotation and flipping.

Model parameters The model used for the experiments consists of 128 feature channels, iterated with 64 steps. The fully connected hidden layer and the classifier have a dimension of 128. The learnable attention map has the same size as the input image, which contains 64×64 pixels.

Training As most of the datasets are imbalanced in their classwise distribution, we use focal loss [10] for training. On the datasets Matek-19, Acevedo-20, INT-20, Sipakmed, Urine-23, and Malaria-18, the model is trained for 32 epochs with a batch size of 16. We use Adam optimizer with a learning rate of 0.0004, a β_1 of 0.9 and β_2 of 0.999 for training with an exponential learning rate decay with weight 0.9999. On the datasets CRC and PatchCam, the model is only trained for 10 and 5 epochs, respectively, because they converge faster with a bigger amount of data. We adapted the learning rate to 0.001 for CRC and 0.004 for PatchCam.

Metrics We perform stratified five fold cross validation and report the accuracy score on the validation split on most of the datasets. On CRC and PatchCam, where the data splits are predefined, we computed the means and standard deviations from five independent runs, and we report balanced accuracy score on CRC to align with the published baseline. The accuracies on PatchCam are evaluated on the test dataset.

Baselines To compare our approach with existing lightweight architectures, we evaluated MobileNetV3 (mobilenetv3_small_050) and MobileViT (mobilevit_xxs) using the TIMM library [26]. In preliminary experiments, we identified 0.003 as an effective learning rate for both models, which was subsequently used across all datasets. The optimizer, batch size, augmentation, and loss function were kept identical to the aNCA training settings to maintain consistency. The only exception was for CRC and PatchCam, where training was limited to 16 epochs due to faster convergence on these larger datasets. We prioritized a consistent training pipeline to ensure a fair comparison, while acknowledging that additional hyperparameter tuning and model-specific optimizations could further improve baseline performance. To support reproducibility, we provide the full codebase, including training and evaluation scripts, along with detailed instructions for reproducing our results at [anonymized link].

3.3 Results

The model's performance is evaluated against WBC-NCA [4], MobileNetV3 [7] (a lightweight CNN), MobileViT$_{\text{xxs}}$ [14] (a lightweight ViT), and published state of the art for every dataset. Table 1 shows the mean and standard deviation of accuracy of the mentioned methods, as well as the number of their parameters.

Table 1. aNCA consistently outperforms lightweight baselines across all datasets. Classification accuracies of our proposed aNCA against WBC-NCA, MobileNetV3, MobileViT$_{\text{xxs}}$, and dataset-specific baselines are reported across eight datasets, with their model sizes in parentheses. Standard accuracy is reported for most datasets, while balanced accuracy is used for CRC to align with the published baseline. The highest accuracy among lightweight models (<1M parameters) for each dataset is highlighted in **bold**.

Dataset \ Method	**aNCA** (89k)	WBC-NCA (86k)	CNN (580k)	ViT (955k)	Published (21M-139M)
Matek-19	**95.3±0.4**	92.6±0.4	90.1±0.3	92.2±0.2	96.1 [13]
Acevedo-20	**92.1±0.3**	89.8±0.7	79.6±0.6	85.6±0.3	96.2 [1]
INT-20	**92.9±0.2**	88.0±0.3	86.3±0.4	89.7±0.3	88.7 [22]
Sipakmed	**94.3±1.1**	92.5±0.7	82.9±2.0	87.8±1.7	99.1 [8]
Urine-23	**92.7±0.4**	87.9±1.3	79.6±1.5	85.4±1.5	96.0 [29]
Malaria-18	**96.7±0.2**	96.5±0.3	93.4±1.7	95.2±0.4	98.9 [18]
CRC	**91.6±0.5**	87.7±1.3	70.5±4.1	74.6±1.8	93.0 [19]
PatchCam	**83.5±0.3**	82.5±0.2	73.6±0.9	73.9±0.8	85.4 [19]

Published Baselines Matek et al. [13] reported an accuracy of 96.1% on Matek-19 with a model based on ResNeXt [27]. Salehi et al. [22] trained the same model on INT-20, reporting an accuracy of 88.7%. Acevedo et al. [1] report an accuracy of 96.2% with a VGG-16 architecture [23] on Acevedo-20. Karri et al. [8] reached an accuracy of 99.12% using a 19-layer CNN on Sipakmed [16]. Muhammed et al. [29] proposed a hybrid model combining textural features with a ResNet50 [5] backbone on Urine-23, achieving 96% accuracy. Rajaraman et al. [18] applied a VGG-16 [23] backbone on the Malaria-18 dataset with 98.9% accuracy. For the two pathology datasets, CRC and PatchCam, we use

ViT-Small proposed by Roth et al. [19] as the baseline. Specifically, we take the reported CRC results from the paper and reproduce the same setup to train on PatchCam. To maintain consistency with their results, we present the balanced accuracy score for CRC, while standard accuracy is used for all other datasets.

3.4 Ablation studies

We conducted ablation studies based on two design choices. Firstly, we consider **different percentages of pixel values** used for averaging in the pooling operation. Our results indicate that selecting the top 10% yields the highest accuracy across most datasets. Table 2 presents the model's performance using different percentages. Secondly, we evaluate an **alternative implementation of the attention map**. In the proposed aNCA model, we define an array of learnable weights used in a weighted average. Additionally, we explore an alternative approach where the attention map is generated by combining the extracted feature maps via a learnable convolutional layer. The last column of Table 2 show its accuracies.

Table 2. Averaging the highest 10% of pixels in feature maps achieves the highest performance on most datasets. For each NCA feature channel, we take the average of the highest x percent of pixel values as the reduced feature. We studied the performance when setting x to 5%, 10%, 20%, and 50% (Column 1-4). We also studied an alternative approach (conv) where the attention map is generated through a convolutional layer that operates on the feature (Column 5). The highest accuracy for each dataset is highlighted in **bold**, the second highest is underlined.

Dataset \ Method	aNCA (5%)	**aNCA (10%)**	aNCA (20%)	aNCA (50%)	aNCA (conv,86k)
Matek-19	95.2±0.3	**95.3±0.4**	**95.3±0.3**	95.1±0.3	94.8±0.3
Acevedo-20	91.8±0.2	**92.1±0.3**	91.6±0.3	91.3±0.3	91.8±0.2
INT-20	92.8±0.2	92.9±0.2	**93.0±0.2**	92.8±0.4	92.2±0.2
Sipakmed	93.1±0.5	**94.3±1.1**	91.9±2.2	92.4±0.8	92.5±0.8
Urine-23	92.0±0.5	**92.7±0.4**	90.9±0.2	89.4±1.2	90.4±0.4
Malaria-18	96.6±0.2	**96.7±0.2**	96.2±0.2	96.3±0.1	96.5±0.3
CRC	91.0±0.9	**91.6±0.5**	91.4±0.3	90.7±0.6	91.1±0.6
PatchCam	83.0±0.4	**83.5±0.3**	82.4±0.6	82.2±0.5	81.3±1.0

4 Conclusion

We introduced aNCA, a novel image classification model that enhances existing NCA models by incorporating attention pooling, resulting in significantly improved accuracy. By learning the optimal spatial weighting through an attention map, aNCA effectively transitions between high-dimensional NCA features and low-dimensional embeddings for the classification head. Additionally, aNCA effectively reduces noisy features by focusing only on the top 10% values of each feature channel.

Evaluations on eight clinical microscopy datasets demonstrate that aNCA consistently achieves high accuracy across various modalities, outperforming both existing NCA models and other lightweight alternatives. Furthermore, aNCA requires substantially fewer parameters compared to conventional methods.

Our work highlights the potential of aNCA in microscopy image classification tasks. As an effective, robust, and explainable feature extractor, it is particularly well-suited for medical settings. While further work is needed to enhance computational efficiency and deployment readiness, our results suggest that NCA-based models may offer a promising alternative for flexible and interpretable approaches to medical image analysis.

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